Assignment No	1
Title	Partitioning
Objective	Range , List And Hash
Roll No	MCA2649

1.Range

Source Code:-

```
CREATE TABLE sales_range
(salesman_id NUMBER(5),
salesman_name VARCHAR2(30),
sales_amount NUMBER(10),
sales_date DATE)
PARTITION BY RANGE(sales_date)
(
PARTITION sales_jan2026 VALUES LESS
THAN(TO_DATE('01/02/2026','DD/MM/YYYY')),
PARTITION sales_feb2026 VALUES LESS
THAN(TO_DATE('01/03/2026','DD/MM/YYYY')),
PARTITION sales_march2026 VALUES LESS
THAN(TO_DATE('01/04/2026','DD/MM/YYYY')),
PARTITION sales_april2026 VALUES LESS
THAN(TO_DATE('01/05/2026','DD/MM/YYYY'))
);
```

Output:-

Selecting the data from the Range partition

```
SQL> select * from sales_range;

SALESMAN_ID SALESMAN_NAME                    SALES_AMOUNT SALES_DAT
----------- ------------------------------ ------------ ---------
          1 John Smith                             5000 26-JAN-26
          5 Shruti Kumari                         15000 01-JAN-26
          2 Satish Rathod                         10000 25-FEB-26
          6 Tejas                                  8000 04-FEB-26
          3 Khushi Chauhan                        15000 20-MAR-26
          7 Anuj                                   8000 06-MAR-26
          4 Vishal CHauhan                         3000 12-APR-26

7 rows selected.
```

```
SQL> SELECT * from sales_range PARTITION(sales_jan2026);

SALESMAN_ID SALESMAN_NAME                    SALES_AMOUNT SALES_DAT
----------- ------------------------------ ------------ ---------
          1 John Smith                             5000 26-JAN-26
          5 Shruti Kumari                         15000 01-JAN-26
```

```
SQL> SELECT * from sales_range PARTITION(sales_feb2026);

SALESMAN_ID SALESMAN_NAME                    SALES_AMOUNT SALES_DAT
----------- ------------------------------ ------------ ---------
          2 Satish Rathod                         10000 25-FEB-26
          6 Tejas                                  8000 04-FEB-26
```

```
SQL> SELECT * from sales_range PARTITION(sales_march2026);

SALESMAN_ID SALESMAN_NAME                    SALES_AMOUNT SALES_DAT
----------- ------------------------------ ------------ ---------
          3 Khushi Chauhan                        15000 20-MAR-26
          7 Anuj                                   8000 06-MAR-26
```

```
SQL> SELECT * from sales_range PARTITION(sales_april2026);

SALESMAN_ID SALESMAN_NAME                  SALES_AMOUNT SALES_DAT
----------- ------------------------------ ------------ ---------
          4 Vishal CHauhan                         3000 12-APR-26
          8 Sidesh                                 9000 06-APR-26
```

```
SQL> SELECT table_name, partition_name, high_value, num_rows
  2  FROM user_tab_partitions
  3  WHERE table_name = 'SALES_RANGE';

TABLE_NAME
--------------------------------------------------------------------------------
PARTITION_NAME
--------------------------------------------------------------------------------
HIGH_VALUE
--------------------------------------------------------------------------------
  NUM_ROWS
----------
SALES_RANGE
SALES_APRIL2026
TO_DATE(' 2026-05-01 00:00:00', 'SYYYY-MM-DD HH24:MI:SS', 'NLS_CALENDAR=GREGORIA
```

```
SQL> ALTER TABLE sales_range TRUNCATE PARTITION sales_jan2026;

Table truncated.

SQL> SELECT * from sales_range PARTITION(sales_jan2026);

no rows selected
```

```
SQL> ALTER TABLE sales_range drop PARTITION sales_jan2026;

Table altered.
```

2.List

Source Code:-

```
CREATE TABLE Student (
   student_id NUMBER(5),
   student_name VARCHAR2(20),
   student_dob DATE)
PARTITION BY LIST (student_name) (
   PARTITION stu_divA VALUES ('a','b','c','d','e','f','g','h','i','j','k'),
   PARTITION stu_divB VALUES ('n','o','p','q','r','s','t','u','v','w','x','y','z'));
```

Output:-

```
SQL> CREATE TABLE Student (
  2       student_id NUMBER(5),
  3       student_name VARCHAR2(20),
  4       student_dob DATE)
  5  PARTITION BY LIST (student_name) (
  6       PARTITION stu_divA VALUES ('a','b','c','d','e','f','g','h','i','j','k'),
  7       PARTITION stu_divB VALUES ('n','o','p','q','r','s','t','u','v','w','x','y'
,'z'));

Table created.
```

```
SQL> select * from student;

STUDENT_ID STUDENT_NAME         STUDENT_D
---------- -------------------- ---------
       221 j                    15-SEP-58
       321 a                    19-AUG-58
       322 d                    19-NOV-58
       323 w                    19-AUG-58
       324 s                    19-JAN-58
       325 t                    19-FEB-58

6 rows selected.
```

```
SQL> select * from student partition(stu_divA);

STUDENT_ID STUDENT_NAME         STUDENT_D
---------- -------------------- ---------
       221 j                    15-SEP-58
       321 a                    19-AUG-58
       322 d                    19-NOV-58
```

```
SQL> select * from student partition(stu_divB);

STUDENT_ID STUDENT_NAME         STUDENT_D
---------- -------------------- ---------
       323 w                    19-AUG-58
       324 s                    19-JAN-58
       325 t                    19-FEB-58
```

```
SQL> alter table student add partition stu_null values(NULL);

Table altered.
```

```
SQL> select * from Student partition(stu_null);

no rows selected
```

```
SQL> SELECT table_name, partition_name, high_value, num_rows
  2  FROM user_tab_partitions
  3  WHERE table_name = 'STUDENT';

TABLE_NAME
------------------------------------------------------------------------------
PARTITION_NAME
------------------------------------------------------------------------------
HIGH_VALUE
------------------------------------------------------------------------------
  NUM_ROWS
----------
STUDENT
STU_DIVA
'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', 'k'
```

```
SQL> alter table Student add partition stu_default values(DEFAULT);

Table altered.

SQL> select * from Student partition(stu_default);

no rows selected
```

3.Hash

Source Code:-

Create table sales

```
 2  (
 3  dept_no NUMBER,
 4  part_no VARCHAR2(20),
 5  country VARCHAR(20),
 6  date1 DATE,
 7  amount NUMBER
 8  )
 9  PARTITION BY HASH(part_no)
10  (
11  PARTITION p1,
12  PARTITION p2,
13  PARTITION p3
14  );
```

Output:-

```
SQL> Create table sales
  2  (
  3  dept_no NUMBER,
  4  part_no VARCHAR2(20),
  5  country VARCHAR(20),
  6  date1 DATE,
  7  amount NUMBER
  8  )
  9  PARTITION BY HASH(part_no)
 10  (
 11  PARTITION p1,
 12  PARTITION p2,
 13  PARTITION p3
 14  );

Table created.
```

```
SQL> INSERT INTO sales VALUES(1,'101','INDIA',TO_DATE('23/02/2026','DD/MM/YYYY'),25000);

1 row created.

SQL> INSERT INTO sales VALUES(2,'102','USA',TO_DATE('15/03/2026','DD/MM/YYYY'),18000);

1 row created.

SQL> INSERT INTO sales VALUES(3,'103','UK',TO_DATE('20/04/2026','DD/MM/YYYY'),32000);

1 row created.

SQL> INSERT INTO sales VALUES(4,'105','CANADA',TO_DATE('12/06/2026','DD/MM/YYYY'),15000);

1 row created.

SQL> INSERT INTO sales VALUES(5,'106','AUSTRALIA',TO_DATE('25/07/2026','DD/MM/YYYY'),41000);

1 row created.
```

```
SQL> SELECT * FROM sales PARTITION(p1);

   DEPT_NO PART_NO              COUNTRY              DATE1         AMOUNT
---------- -------------------- -------------------- --------- ----------
         2 102                  USA                  15-MAR-26      18000
         5 106                  AUSTRALIA            25-JUL-26      41000
```

```
SQL> SELECT * FROM sales PARTITION(p2);

   DEPT_NO PART_NO              COUNTRY              DATE1         AMOUNT
---------- -------------------- -------------------- --------- ----------
         3 103                  UK                   20-APR-26      32000
```

```
SQL> SELECT * FROM sales PARTITION(p3);

   DEPT_NO PART_NO              COUNTRY              DATE1         AMOUNT
---------- -------------------- -------------------- --------- ----------
         1 101                  INDIA                23-FEB-26      25000
         4 105                  CANADA               12-JUN-26      15000
```